solving standard divisibilities

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standard tricks:

1) simplify the multiple

$n \mid \cancel{n^2}+4 \to n \mid 4$

$n \mid n^2+4 \Rightarrow n \mid n^2+4n+4 = (n+2)^2$

2) bounding ↗ abs values!!

- $a \mid b \longrightarrow |a| \leq |b|$ or $b=0$
- b/a, e.g. $b/a < 3$, $b,a>0$ $\Rightarrow b=a$ or $b=2a$

1. (AIME 1986 Q5) What is the largest positive integer n for which n^3+100 is divisible by $n+10$?

$n+10 \mid n^3+100$

$\Rightarrow n+10 \mid n^3+100+30n(n+10)$
$= n^3+100+30n^2+300n$
$= (n+10)^3-900$

$\Rightarrow n+10 \mid 900$

$n+10=900 \Rightarrow \boxed{n=890}$

$n+10 \mid n^3+100-(n^3+1000)$
$\Rightarrow n+10 \mid 900$

2. (ELMO SL 2024 N1) Find all pairs (n,d) of positive integers such that $d \mid n^2$ and $(n-d)^2 < 2d$.

$d \mid n^2$
$\Rightarrow d \mid n^2 - 2nd^2 + d^2$
$\Rightarrow d \mid (n-d)^2$

$(n-d)^2 < 2d$

if $n=d$,

if $n \neq d$,

if $n=d$,
it works
(a,a)

if $n\neq d$,
$(\underbrace{n-d}_{k})^2=d$
$\Rightarrow k^2=n-k$
$\Rightarrow n=k^2+k$
$d=k^2$
(k^2+k,k^2)

3. (JBMO 2013 P1) Find all ordered pairs (a,b) of positive integers for which the numbers $\dfrac{a^3b-1}{a+1}$ and $\dfrac{b^3a+1}{b-1}$ are both positive integers.

$a+1 \mid a^3b-1$
$\Rightarrow a+1 \mid a^3b-1-a^2b(a+1)$
$=-a^2b-1$
$\Rightarrow a+1 \mid a^2b+1-ab(a+1)$
$=1-ab$
$\Rightarrow a+1 \mid 1-ab+b(a+1)$
$=1+b$
$\Rightarrow a+1 \mid b+1$

$b-1 \mid b^3a+1$
$\Rightarrow b-1 \mid b^3a+1-b^2a(b-1)$
$=b^2a+1$
$\Rightarrow b-1 \mid b^2a+1-ab(b-1)$
$=ab+1$
$\Rightarrow b-1 \mid ab+1-a(b-1)$
$=a+1$
$\Rightarrow b-1 \mid a+1$

$\therefore b-1 \mid b+1 \Rightarrow b-1 \mid 2 \rightarrow b=2 \rightarrow a+1 \mid 3$
$\Rightarrow a=2$
$\rightarrow b=3 \rightarrow a+1 \mid 4$
$\Rightarrow a=1,3$

$\bigstar$ $x-y \mid x^n-y^n \ \forall\ n\in\mathbb{N}$
$x-y\equiv 0 \pmod{x-y}$
$x\equiv y \pmod{x-y}$
$x^n\equiv y^n \pmod{x-y}$
$x^n-y^n\equiv 0 \pmod{x-y}$

$x+y \mid x^n+y^n$ for all odd n

$$a+1 \mid a^3b-1-b(a^3+1) = -1-b$$
$$\Rightarrow a+1 \mid b+1$$

4. (IMOSL 2021 N1) Find all positive integers $n \geq 1$ such that there exists a pair (a, b) of positive integers, such that a^2+b+3 is not divisible by the cube of any prime, and

$$n = \frac{ab+3b+8}{a^2+b+3}.$$

$$a^2+b+3 \mid ab+3b+8$$
$$\Rightarrow a^2+b+3 \mid ab+3b+8-(a+3)(a^2+b+3)$$
$$= -a^3-3a^2-3a-9+8$$
$$= -a^3-3a^2-3a-1$$
$$= -(a+1)^3$$

$$\Rightarrow a^2+b+3 \mid (a+1)^3 \qquad p^3 \nmid a^2+b+3$$
$$\Rightarrow a^2+b+3 \mid (a+1)^2 \qquad \text{or } v_p(a^2+b+3) \leq 2$$

for p s.t. $p \mid a^2+b+3$,

$p \mid a+1$

$v_p((a+1)^2) \geq 2 \geq v_p(a^2+b+3)$

$$a^2+b+3 \mid a^2+2a+1-(a^2+b+3)$$
$$= 2a-b-2$$

case 1: $2a-b-2>0$

$2a-b-2 \geq a^2+b+3$

$2a-a^2 \geq 2b+5>5$

$\rightarrow\leftarrow$

case 2: $2a-b-2=0$

$b=2a-2 \qquad (a,b)=(2,2)$

$$n=\frac{ab+3b+8}{a^2+b+3} = \frac{a(2a-2)+3(2a-2)+8}{a^2+(2a-2)+3} = 2$$

case 3: $2a-b-2<0$

$a^2+b+3 \mid b+2-2a$

$\cancel{b}+2-2a \geq a^2+\cancel{b}+3$

$0 \geq a^2+2a+1$

$= (a+1)^2$

$a=-1 \qquad \rightarrow\leftarrow$

$\therefore n = 2$

5. (Iran MO R2 2005 P1) Let $n, p > 1$ be positive integers and p be prime. We know that $n \mid p-1$ and $p \mid n^3 - 1$. Prove that $4p - 3$ is a perfect square.

$n \mid p-1$

$\downarrow$

$p = kn+1$

$p \mid n^3 - 1 = (n-1)(n^2+n+1)$

$p \nmid n-1$ since $p = kn+1 > n-1$

so $p \mid n^2+n+1$

$kn+1 \mid n^2+n+1$

$kn+1 \mid n^2+n+1-(kn+1)$

$= n^2+n-kn$

$= n(n+1-k)$ $\quad \gcd(n, kn+1) = 1$

diff of cubes:

$a^3 - b^3 = (a-b)(a^2+ab+b^2)$

sum of cubes:

$a^3 + b^3 = (a+b)(a^2-ab+b^2)$

$kn+1 \mid n+1-k$

$\searrow$ $n+1-k = 0$

$\boxed{k = n+1}$

$\downarrow$

$|n+1-k| \geq kn+1$

$\swarrow$

$n + \cancel{1} - k \geq kn + \cancel{1}$

$n \geq kn + k$

$\rightarrow\leftarrow$

$\searrow$

$-(n+1-k) \geq kn+1$

$-n-1+k \geq kn+1$

$k \geq kn+n+2$

$\rightarrow\leftarrow$

$p = kn+1 = (n+1)n+1 = n^2+n+1$

$4p-3 = 4(n^2+n+1) - 3 = (2n+1)^2$ ■

6. (APMO 2022 P1) Find all pairs (a, b) of positive integers such that a^3 is multiple of b^2 and $b-1$ is multiple of $a-1$.

$b^2 \mid a^3$

$a-1 \mid b-1$

works:

(a, a) , $(a, 1)$

$a \neq 1$

$b=1$ works for all a

$a^3 = kb^2$ (nice, no addtn/subtrctn)

$a = dm$, $b = dn$, $d = \gcd(a,b)$

$\gcd(m,n) = 1$

$\rightarrow d^3m^3 = kd^2n^2$

$dm^3 = kn^2$

$d = cn^2$, $k = cm^3$

$\rightarrow a = cn^2m$, $b = cn^3$

$cn^2m - 1 \mid cn^3 - 1$

$\Rightarrow cn^2m - 1 \mid cn^3 - 1 - (cn^2m - 1)$

$= cn^3 - cn^2m$

$= cn^2(n-m)$

$\gcd(cn^2m - 1, cn^2) = 1$

$\Rightarrow cn^2m - 1 \mid n - m$

$n - m = 0$

$\Rightarrow n = m$

$\Rightarrow a = b$ ✓

$|n-m| \geq cn^2m - 1$

$n - m \geq cn^2m - 1$

$n + 1 \geq cn^2m + m$

recall that $c, m, n \in \mathbb{Z}^+$

$c = n = m = 1$

$\Rightarrow (a,b) = (1,1) \rightarrow \leftarrow$

$m - n \geq cn^2m - 1$

$0 \geq 1 - n \geq m(cn^2 - 1) \rightarrow c = n = 1$

$m =$ anything

$\therefore (a, b) = (a, 1)$ ✓

7. (APMO 2013 P2) Determine all positive integers n for which

$$\frac{n^2+1}{\lfloor\sqrt{n}\rfloor^2+2}$$

is an integer.

$$\lfloor\sqrt{n}\rfloor^2+2 \mid n^2+1$$

$$n = k^2+m, \quad 0 \le m \le 2k$$

$$k^2+2 \mid (k^2+m)^2+1$$

$$k^2+2 \mid k^4+2k^2m+m^2+1$$

$$k^2+2 \mid k^4+2k^2m+m^2+1-(k^2+2)(k^2-2)$$
$$= 2k^2m+m^2+5$$

$$k^2+2 \mid 2k^2m+m^2+5-2m(k^2+2)$$
$$= m^2-4m+5 = (m-2)^2+1 > 0$$

$$\frac{m^2-4m+5}{k^2+2} < 4 \quad \text{since } m \le 2k$$

(1) $m^2-4m+5 = k^2+2$
$\Rightarrow (m-2)^2+1 = k^2+2$
$\Rightarrow (m-2)^2 = k^2+1$
$\Rightarrow k=0,\ m-2=\pm 1$
$\rightarrow\leftarrow$

(2) $(m-2)^2+1 = 2(k^2+2)$
$m-2 \equiv k \equiv 0 \pmod 3$
$9 \mid (m-2)^2-2k^2$
$(m-2)^2+1 = 2k^2+4$
$\rightarrow\leftarrow$ mod 9

(3) $(m-2)^2+1 = 3(k^2+2)$
$\rightarrow\leftarrow$ mod 3

no solutions!!

8. (APMO 2002 P2) Find all positive integers a and b such that

$$\frac{a^2+b}{b^2-a} \quad \text{and} \quad \frac{b^2+a}{a^2-b}$$

are both integers.

symmetric $\Rightarrow$ assume order WLOG

assume $a \geq b$

$a = b$

$a^2 - a \mid a^2 + a - (a^2 - a)$

$a^2 - a \mid 2a$

$\cancel{a}(a-1) \mid 2\cancel{a}$

$a - 1 \mid 2$

$a = 2, 3$

$\therefore (2,2), (3,3)$

$a > b$

$a = b + n$

$\underbrace{(b+n)^2 - b}_{+} \mid \underbrace{b^2 + b + n}_{+}$

if $n > 1$,

$(b+n)^2 - b = b^2 + (2n-1)b + n^2$
$> b^2 + b + n$

$\therefore n = 1 \Rightarrow a = b + 1$

* fulfills 2nd condition

1st condition:

$b^2 - (b+1) \mid (b+1)^2 + b$
$= b^2 + 3b + 1$

$b^2 - b - 1 \mid b^2 + 3b + 1 - (b^2 - b - 1)$
$= 4b + 2$
$= 2(2b+1)$

$b^2 - b - 1 \mid 2b + 1$

$b = 1$
$a = 2$ ✓

$b > 1$

$b^2 - b - 1 \leq 2b + 1$

$b^2 \leq 3b + 2 \Rightarrow b < 4$

$b = 2$
$1 \mid 5$ ✓

$b = 3$
$5 \mid 7$ ✗

$b=2$ $b=3$

$1 \mid 5$ ✓ $5 \mid 7$ ✗

$a=3$

$\therefore (2,2), (3,3), (2,1), (1,2),$
$(3,2), (2,3)$

9. (ELMO SL 2024 N4) Find all pairs (a, b) of positive integers such that $a^2 \mid b^3 + 1$ and $b^2 \mid a^3 + 1$.

assume WLOG $a \geq b$

$a^2 \mid (b+1)(b^2-b+1)$??

trick: equate the multiple!!

$a^2 \mid a^3+b^3+1$ $\quad\quad \gcd(a^2, b^2)=1$

$b^2 \mid a^3+b^3+1$

$\Rightarrow a^2b^2 \mid a^3+b^3+1$

$$a^2b^2 \leq a^3+b^3+1 \leq 2a^3+1$$

$$b^2 \leq 2a + 1/a^2 \leq 2a+1$$

$$a^2 \leq b^3+1$$

$$\frac{b^2-1}{2} \leq a \Rightarrow \frac{b^4-2b^2+1}{4} \leq b^3+1$$

$$b^4-2b^2+1 \leq 4b^3+4$$

$$b^4 \leq 4b^3+2b^2+3$$

but $4b^3+2b^2+3 > b^4$ if $b \geq 4$

so $b=1, 2$ or 3

$b=1 \Rightarrow a=1$

$b=2 \Rightarrow a=3$

$b=3 \Rightarrow a=2$

10. (IMO 1998 P4) Determine all pairs (x, y) of positive integers such that $x^2y + x + y$ is divisible by $xy^2 + y + 7$.

$$xy^2+y+7 \mid x^2y+x+y$$

$$xy^2+y+7 \mid x^2y^2+xy+y^2 - x(xy^2+y+7)$$

$$= y^2 - 7x$$

$$y^2 - 7x > 0$$

$$y^2 - 7x \geq xy^2+y+7$$

$\rightarrow\leftarrow$

$$y^2 - 7x = 0$$

$$y^2 = 7x$$

$$(x,y) = (7k^2, 7k)$$

works for all $k \in \mathbb{N}$

$$y^2 - 7x < 0$$

$$7x > 7x - y^2 \geq xy^2+y+7 > xy^2$$

$\underline{y=1}$ or $\underline{2}$

$$x+8 \mid x^2+x+1$$

$$x+8 \mid x^2+x+1 - x(x+8)$$

$$= 1-7x$$

$$x+8 \mid 1-7x+7(x+8)$$

$$= 57$$

$$x = 11, 49$$

$$4x+9 \mid 2x^2+x+2$$

$$4x+9 \mid 4x^2+2x+4$$

$$- x(4x+9)$$

$$= 4-7x$$

$$4x+9 \mid 4-7x+2(4x+9)$$

$$= x+22$$

$$x+22 \geq 4x+9$$

$$13 \geq 3x$$

$$x = 1,2,3,4$$

$\rightarrow\leftarrow$